

Problem

Draw the Bode plots for the transfer function

$$H(\omega) = \frac{5(j\omega + 2)}{j\omega(j\omega + 10)}$$

Step-by-step solution

Step 1 of 6

Given transfer function is,

$$H(\omega) = \frac{5(j\omega + 2)}{j\omega(j\omega + 10)}$$

$$H(\omega) = \frac{5 \times 2 \left(1 + \frac{j\omega}{2}\right)}{10j\omega \left(1 + \frac{j\omega}{10}\right)}$$

$$\therefore H(\omega) = \frac{1 + \frac{j\omega}{2}}{j\omega \left(1 + \frac{j\omega}{10}\right)}$$

$$\Rightarrow H(\omega) = \frac{\left|1 + \frac{j\omega}{2}\right|}{\left|j\omega\right| \left|1 + \frac{j\omega}{10}\right|} \angle \left[90^\circ + \tan^{-1}\left(\frac{\omega}{2}\right) - \tan^{-1}\left(\frac{\omega}{10}\right)\right]$$

Hence, the magnitude and phase are,

$$H_{dB} = 20 \log_{10} \left(\sqrt{1 + \left(\frac{\omega}{2}\right)^2} \right) - 20 \log_{10}(\omega) - 20 \log_{10} \left(\sqrt{1 + \left(\frac{\omega}{10}\right)^2} \right) \dots\dots\dots (1)$$

$$\phi = -90^\circ + \tan^{-1}\left(\frac{\omega}{2}\right) - \tan^{-1}\left(\frac{\omega}{10}\right) \dots\dots\dots (2)$$

The various terms of $H(\omega)$ are listed in the increasing order of their corner frequencies. Also the table shows the slope contributed by each term and the change in slope at the corner frequency.

Step 2 of 6

Term	Corner Frequency (rad/sec)	Slope db/decade	Change in slope db/decade
$\frac{1}{j\omega}$	—	—20	—
$1 + \frac{j\omega}{2}$	$\omega = 2$	+20	$-20 + 20 = 0$
$\frac{1}{1 + \frac{j\omega}{10}}$	$\omega = 10$	-20	$0 - 20 = -20$

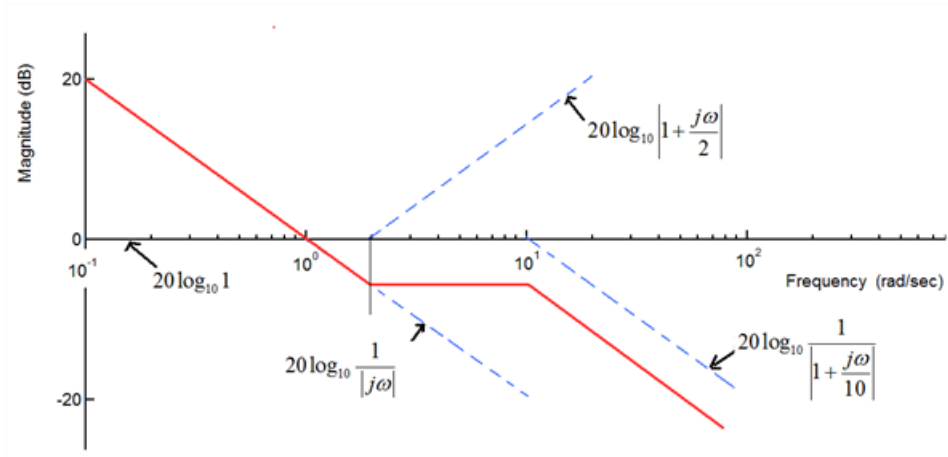
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Below table shows gain values for different values of ‘ ω ’ according to equation (1)

ω	H_{dB}
0.1	20
2	-3.2
10	-8.9

The magnitude plot is

Step 4 of 6



Step 5 of 6

The phase angle of ϕ are calculated for various values ω and listed in below

$$\phi = -90^\circ + \tan^{-1} \left(\frac{\omega}{2} \right) - \tan^{-1} \left(\frac{\omega}{10} \right)$$

ω	$\phi = \angle H(\omega) \text{ (Deg)}$
1	-69.2
20	-69.2

The phase plot is

Step 6 of 6

